Set 26: Gas volumes

Notes

From our understanding of kinetic theory we know that gas volumes are independent of the identity of the gas. Gas volume is dependent upon the number of particles, its pressure and volume. This means that, for an ideal gas, one mole of any gas will occupy the same volume under the same pressure and temperature conditions. This enables us to develop a relationship between volume and number of moles of a gas.

Molar volume of gases

The molar volume of a gas is the volume occupied by 1 mole of the gas. The molar volume for an ideal gas, and for many real gases, is 22.71 L at S.T.P.
The relationship between the number of moles (n) of a gas and its volume (V) in litres at S.T.P. is

$$n = \frac{V(\text{in litres at S.T.P.})}{22.71}$$

Examples

1. Calculate the volume occupied by 2.75 moles of carbon monoxide at S.T.P.

$$n = \frac{V}{22.71}$$

$$V = n \times 22.71$$
$$= (2.75)(22.71)$$
$$V = 62.5 \text{ L}$$

2. What is the volume of 9.60 g of oxygen at S.T.P.?

(a) Calculate the number of moles.

$$n(O_2) = \frac{9.60}{32.0} \qquad M(O_2) = 32.0 \text{ g mol}^{-1}$$

$$= 0.300 \text{ mol}$$

(b) Find the volume at S.T.P

$$n = \frac{V}{22.71}$$

$$V = n \times 22.71$$
$$= (0.300)(22.71)$$
$$V = 6.81 \text{ L at S.T.P.}$$

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Notes

Set 26: Exercises

1. Calculate the volume occupied by:
   (a) 6.50 moles of carbon dioxide at S.T.P.
   (b) 0.850 moles of hydrogen at S.T.P.

2. Calculate the number of moles of:
   (a) methane (CH_4) in 4.50 L of the gas at S.T.P.
   (b) oxygen in 25.0 mL of the gas at S.T.P.

3. Calculate the volume of
   (a) 100 g of propane (C_3H_8) at S.T.P.
   (b) carbon dioxide gas at S.T.P. produced when a 1.00 kg block of dry ice, which is solid CO_2, sublimes.

4. Calculate the volume occupied by the following masses of gases at S.T.P.
   (a) 1.34 g of methane (CH_4)
   (b) 2.00 g of ethane (C_2H_6)
   (c) 58.6 g of ammonia (NH_3)
   (d) 0.566 g of sulfur dioxide (SO_2)

5. 4.18 g of a gas occupies 1.00 L at S.T.P. Calculate the molar mass of the gas.

6. 3.49 g of a gas occupies 1.00 L at S.T.P. Calculate the molar mass of the gas.

7. What mass of oxygen gas will occupy 1.00 L at S.T.P.?

8. What masses will the following volumes of gas weigh?
   (a) 2.55 L of sulfur trioxide (SO_3)
   (b) 89.5 L of nitrogen (N_2)
   (c) 0.00253 L of argon
   (d) 41.2 L of chlorine ($C\ell_2$)